

04_sensitivity_K

February 16, 2026

1 04 — Sensitivity to Rashomon size K

Purpose: Test sensitivity to K (Rashomon size). Vary K {5, 10, 15, 25}, compute multiplicity and spatial metrics, plot stabilization curves.

1.1 Imports and path setup

```
[9]: import sys
from pathlib import Path

ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))
if str(ROOT / "src") not in sys.path:
    sys.path.insert(0, str(ROOT / "src"))

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from analysis.experiment_runner import _get_run_dirs
from analysis.preprocessing import get_transformed_test_features
from analysis.run_analysis import load_meta, run_multiplicity, run_spatial

RESULTS_DIR = ROOT / "results"
```

1.2 Loop over K and store results

```
[10]: DATASET = "compas"
K_LIST = [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
k_nn = 30

dataset_dir = RESULTS_DIR / DATASET
run_dirs = _get_run_dirs(dataset_dir)

def run_one_k(run_dir, X_test, K):
    n_cand = len(load_meta(run_dir))
    if K > n_cand:
```

```

        return None
    K_actual = min(K, n_cand)
    mult = run_multiplicity(run_dir, K=K_actual)
    spatial = run_spatial(run_dir, X_test, K=K_actual, k=k_nn)
    return {"mean_variance": mult["mean_variance"], "moran_i":
    ↪spatial["moran_i"], "n_hh": int(np.sum(spatial["HH_mask"]))}

results_k = []
for K in K_LIST:
    for run_dir in run_dirs:
        X_test = get_transformed_test_features(run_dir, DATASET)
        res = run_one_k(run_dir, X_test, K)
        if res is not None:
            results_k.append({"K": K, **res})

df_k = pd.DataFrame(results_k)
print(df_k.groupby("K").agg({"mean_variance": ["mean", "std"], "moran_i":
    ↪["mean", "std"], "n_hh": "mean"}).round(4))

```

	mean_variance		moran_i		n_hh
	mean	std	mean	std	mean
K					
5	0.0005	0.0003	0.2010	0.0927	89.3
10	0.0008	0.0004	0.1768	0.0875	76.7
15	0.0010	0.0004	0.1917	0.0795	91.0
20	0.0011	0.0003	0.1955	0.0674	86.2
25	0.0013	0.0003	0.1993	0.0772	80.7
30	0.0015	0.0003	0.1970	0.0726	80.4
35	0.0016	0.0002	0.1922	0.0626	86.8
40	0.0018	0.0002	0.1933	0.0598	85.2
45	0.0019	0.0002	0.1997	0.0597	84.6
50	0.0020	0.0002	0.2112	0.0600	87.4

1.3 Plot elbow curves

```

[11]: agg_k = df_k.groupby("K").agg(
    mean_variance_mean=("mean_variance", "mean"),
    ↪mean_variance_std=("mean_variance", "std"),
    moran_mean=("moran_i", "mean"), moran_std=("moran_i", "std"),
    n_hh_mean=("n_hh", "mean")).reset_index()

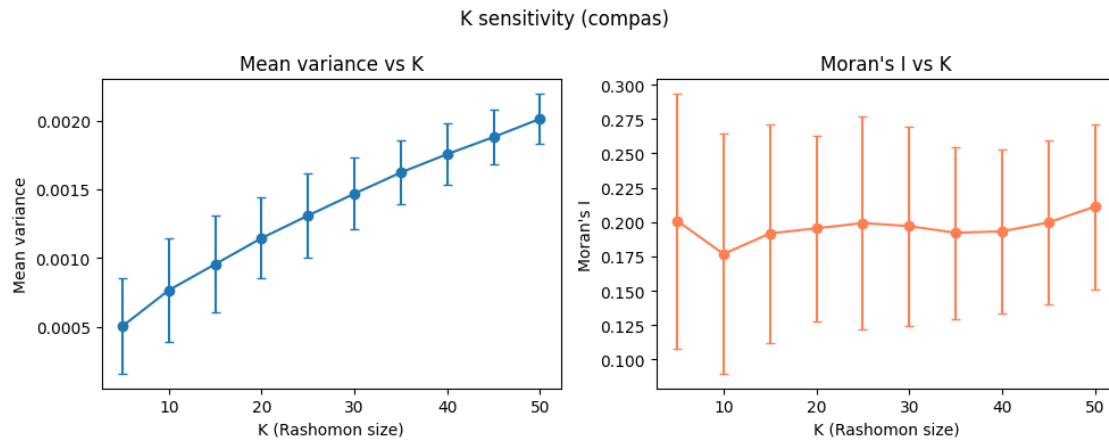
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
ax1.errorbar(agg_k["K"], agg_k["mean_variance_mean"],
    ↪yerr=agg_k["mean_variance_std"], marker="o", capsize=3)
ax1.set_xlabel("K (Rashomon size)")
ax1.set_ylabel("Mean variance")
ax1.set_title("Mean variance vs K")

```

```

ax2.errorbar(agg_k["K"], agg_k["moran_mean"], yerr=agg_k["moran_std"],
            marker="o", capsize=3, color="coral")
ax2.set_xlabel("K (Rashomon size)")
ax2.set_ylabel("Moran's I")
ax2.set_title("Moran's I vs K")
plt.suptitle(f"K sensitivity ({DATASET})")
plt.tight_layout()
plt.show()

```



1.4 Interpret stability

If mean variance and Moran's I stabilize (flat or elbow) as K increases, the chosen K is not driving artifacts.